The Invoice App
Database Architecture & Schema

Revision 2 – corrected to match shipped implementation

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Overview

This document outlines the relational database structure for the Invoice App. Instead of utilizing multiple floating spreadsheet files, the system uses a single SQLite database (`MasterDB.db`) containing four interconnected tables. This guarantees data integrity, prevents duplicate entries, and allows for complex financial reporting (like tracking total profits while ignoring cancelled orders).

What changed since Revision 1 (June 24, 2026). The original schema document was written before the stock-tracking design was finalized and described `stock_quantity` as defaulting to `0`. The shipped implementation instead uses `NULL` to mean "untracked" and `0` to mean "genuinely empty" – a deliberate distinction, not a default-value convenience. This revision corrects that field description; no other schema details changed.

1 Products (The Inventory / Menu)

Functionality: This table acts as the central menu. It holds all available items for sale, their respective retail and wholesale pricing, and an optional stock counter. It seamlessly absorbs data directly from the original Excel/CSV price catalog while preserving legacy product IDs.

Field Name	**Data Type**	**Description**
Product_ID	INTEGER (PK)	Unique identifier for each item. Explicitly matches the original Excel/CSV `prod_ID`.
item_name	TEXT	The primary name or description of the product.
description	TEXT	Additional details about the item (currently mirrors item_name, since the source catalog has no separate name field).
Retail_Price	REAL	The standard retail price of the item.
Wholesale_Price	REAL	The discounted or bulk price of the item.

Field Name	Data Type	Description
stock_quantity	**INTEGER**	**Corrected:** `NULL` means this product's stock is not tracked at all – it never triggers a low-stock or oversell warning, regardless of quantity sold. `0` means the product genuinely has zero units in stock, which *does* trigger warnings on sale. This distinction was introduced specifically because using `0` for both cases made "untracked" and "actually empty" indistinguishable in dashboard stats and sale-time warnings.
Cost	REAL DEFAULT 0.0	The baseline cost of the item, used to compute per-sale profit.

2 Customers (The Client Book)

Functionality: Keeps track of everyone who buys from you. This prevents having to type out contact information manually for every invoice and remembers whether a specific customer should automatically receive wholesale or retail pricing.

Field Name	Data Type	Description
customer_id	INTEGER (PK)	Unique, auto-incrementing ID for the customer.
Name	TEXT NOT NULL	The full name of the customer or business.
Phone_Number	TEXT	Contact information for the client.
Default_Tier	TEXT	Automatically flags the customer as 'retail' or 'wholesale' to quickly apply the correct pricing matrix. Always overridable per-invoice; an override does not change the stored default.

3 Orders (The Invoice Headers)

Functionality: This is the financial core. It represents the actual "bill" itself. It stores the grand totals, the date of purchase, and the overall profit. Crucially, it features a status column that allows for reversibility – meaning an invoice can be "cancelled" without deleting the record entirely, keeping invoice numbers perfectly sequential.

Field Name	Data Type	Description
Invoice_Number	INTEGER (PK) AUTOINCREMENT	The official invoice number.
Customer_ID	INTEGER NOT NULL (FK)	Links directly to the Customers table.
Date	TEXT NOT NULL	Timestamp of when the invoice was generated.

Field Name	Data Type	Description
Subtotal	REAL NOT NULL	Total amount before any cart-level discount.
Discount	REAL NOT NULL	Overall discount applied to the entire invoice.
Total	REAL NOT NULL	The final amount the customer actually owes.
Profit	REAL NOT NULL	Calculated profit margin for the order. Zeroed out (not deleted) if the order is later cancelled, so profit totals stay accurate without losing history.
Status	TEXT NOT NULL	Defaults to 'active'. Can be changed to 'cancelled' to reverse the order without deleting data.

4 OrderDetails

Functionality: Since a single order can contain many different items, this table breaks down the invoice line by line. It links the master Orders table to the Products table, recording exactly what was bought, how many, and the exact price charged at that moment in time – protecting historical invoices if prices change later.

Field Name	Data Type	Description
Invoice_Number	INTEGER NOT NULL (FK)	Links this line item to its parent invoice/order.
Item_ID	INTEGER NOT NULL (FK)	Links back to the Products table to identify what was sold.
Quantity	INTEGER NOT NULL	The number of units purchased for this specific item.
Price_Sold	REAL NOT NULL	The actual price charged at the time of sale. Deliberately denormalized – never recalculated from the product's current price – so historical invoices stay accurate after prices change.